COL 7364/764 — Assignment 1: Sparse Retrieval

A from-scratch inverted index with BM25+ and RM3

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1 Indexing design

1.1 Analysis chain

The competition engine analyses documents and queries with the same function (`each_surface` in `analysis.cpp`):

- **Tokenisation.** Lowercase; split on any non-alphanumeric character.
- **Dotted initials.** Single letters separated only by '.' are glued (`U.S.` → `us`). A leftover length-1 letter is dropped.
- **Stopwords.** Lucene-style English plus question words (`INDEX_STOPWORDS`) at index and query time. A Fox-style list (`STOPWORDS`) is used only when pruning the RM3 CSR on $N \geq 20{,}000$.
- **Stemming.** Porter by default (own `PorterStemmer`); `IR_STEMMER=1|0` selects S-stem or none. Stems are memoised per thread.

Stemmer and stop flags are stored in uncompressed `meta.bin` (`INDEX_VERSION` = 14) and re-read at load.

The required Boolean/VSM/Python-BM25 path is a *separate* indexer (`indexer.py`): S-stem, no stop list, no acronym folding. Unit tests (`tests/test_retrievers.py`) score that path. Table 1 therefore compares two analysis chains on the same topics, which is what the assignment asks (independently verifiable Boolean/VSM vs. BM25), not a single tokenizer.

1.2 What is persisted

Five files reconstruct the scorer. Four of them may be gzipped (`postings.bin` level 6, others 9) if gzip is smaller; `meta.bin` is left plain.

- `dict.bin` — sorted terms, front-coded (vbyte prefix + suffix); one vbyte packs df, codec id and `has_title`; postings offset as a vbyte delta. Term ids are array positions (binary search).
- `postings.bin` — $df < 8$: gap+tf vbyte with a flag for (tf=1). Longer lists: Simple-9 on $tf - 1$ plus Elias–Fano on doc ids, or `CODEC_S9EF_UNIT` if every body tf is 1 (no Simple-9 payload). Title tf is a sparse vbyte sidecar. Long lists are never stored as vbyte: decode of high-df query terms dominated latency.
- `docmap.bin` — if all external ids share a length and a ≤ 64-symbol alphabet, packed base-A codes when that is smaller than raw bytes; equal-length ids otherwise stored raw; variable-length ids front-coded.
- `doclens.bin` — N, avgdl, corpus token total, and one byte $|d|/\text{avgdl} = 64 \log_2(1 + r)$; code 255 saturates at 64.

Not on disk: RSJ IDF (closed form at load), collection frequency and the RM3 CSR (rebuilt in `load_index`; the harness times build, not load), and positions. Postings gzip is four concatenated members (caller compresses chunk 0). `mtime`=0. Load inflates `.gz` members to a temp dir and `mmap`s.

1.3 Build path

`build_index_from_file` `mmap`s JSONL, splits line spans, and runs SPIMI. The parser unescapes only when a backslash is present. $W \leq 4$ workers (reduced at 400k/800k docs) own contiguous doc-id ranges. Vocabulary ranges merge/encode in parallel; dictionary front-coding is serial, so W=1 and W=4 write identical files. The caller is worker 0.

2 Retrieval models

2.1 Boolean and vector-space (required)

`boolean_search` is AND/OR over posting sets, unranked, sorted doc ids. Correctness is the unit tests in `tests/test_retrievers.py` (known tiny corpus, hand-checked scores), not nDCG@10. On the 50 CORD-19 topics the Python indexer gives mean $|Q_{\text{AND}}|$=4 with recall 0.004 (34/50 empty conjunctions) versus mean $|Q_{\text{OR}}|$=1.5×10^5 with recall 0.985: AND is too strict for long NL topics, OR is almost the collection. Ranked metrics on a lexicographic prefix of OR are undefined as a retriever comparison, so Table 1 omits them.

`vsm_score` is `lnc.ltc` cosine: $w = (1 + \log tf) \cdot \ln(N/df)$, with $\|d\|$ the full-document L2 norm from build time. Query-subspace norms are not cosine. Ties: score desc, then doc id asc.

2.2 BM25+ (required)

Python `bm25.py` and the C++ inner loop implement the same Okapi/BM25+ body (tunable k_1, b; δ=0 is textbook BM25). C++ additionally normalises query tf and adds a title indicator:

$$w_q \left[\frac{tf_{\text{eff}}(k_1+1)}{tf_{\text{eff}} + k_1(1-b+b\,|d|/\text{avgdl})} + \delta + \beta \mathbb{1}[tf^{\text{title}} > 0] \right], \quad w_q = \frac{qtf_q}{\sum qtf} \ln\left(1 + \frac{N - df + 0.5}{df + 0.5}\right).$$

tf_{eff} is body tf (w_{title}=0 in the shipped config). The title window is the first 12 token *positions* after `each_surface` (a stopword occupies a slot but is not stored). TAT into a generation-stamped accumulator; $k_1(1-b+b\,|d|/\text{avgdl})$ is a 256-entry LUT on the length byte.

2.3 Custom scorer and RM3 (competition)

`custom_scorer.py` min–max fuses Python BM25 (k_1=0.9, b=0.4) and cosine at 0.85/0.15 over a pool of 200. On CORD-19 it is *worse* than Python BM25 (Table 1), so `retrieve()` does not call it.

Competition ranking is C++ BM25+ with optional RM3. From the top *fb_docs* of pass 1, $P(w|R) = \sum_i P(d_i|R)\,(tf_{w,d_i} + \mu P(w|C))/(|d_i| + \mu)$, IDF-weighted, top *fb_terms* mixed as $\alpha \cdot \text{original} + (1-\alpha) \cdot \text{expansion}$. $P(d|R) = \lambda\,\text{uniform} + (1-\lambda)\,\text{log1p(score)}$ with λ=1 on newswire (full second BM25 pass) and λ=0.25 on abstracts (rerank the first-pass pool, capped at 250). Skip expansion if feedback-score variance $< 10^{-6}$; require support ≥ 2 when *fb_docs* ≥ 12. The CSR is per-doc $tf \cdot \text{idf}$ pruned (df band).

2.4 Collection-adaptive parameters

`adapt_params`(N, avgdl) — never query text — interpolates a short-doc endpoint toward a newswire endpoint, then applies regimes (near-duplicate titles, short answers, forums, large abstracts, small full-text, newswire). C++ `retrieve` then overlays query length: RM3/δ/title off for document-as-query; α tilts on very short/long queries. `retrieve_bm25` sets `apply_qlen_heuristics=false`.

The discrete cuts are on *shape*, not on a named BEIR id, and they sit on top of a continuous SHORT→NEWS interpolation (t from analysed avgdl $\in [120, 155]$). An unseen (N, avgdl) that misses every if-branch keeps that blend with RM3 off — it does not look up a fingerprint. On the CORD-19 index I re-scored the 50 topics after lying to `adapt_params` about N and avgdl: ±15% around the true pair (171,332, 109.7) stays in the large-abstract regime (nDCG@10 0.7458, or 0.7317 when +15% avgdl starts the b-blend). Crossing a cut on purpose — avgdl=136 or N=99,999 — falls back to the blend, RM3 off, and nDCG@10 drops to 0.69, still above textbook BM25 (0.636). Table 3 is the same check across real collections: a knob shipped only if it did not collapse a differently shaped set (RM3 stays off on Touché-length forums).

3 Evaluation

All CORD-19 numbers below were re-run for this report (4 Sep 2026) with `harness/metrics.py` (graded nDCG, $2^{\text{rel}} - 1$; MAP/MRR/P@10 binarise rel > 0) on the 50 released TREC-COVID topics (N=171,332, analysed avgdl=109.71). Held-out topics were never used. C++ rows share Porter+stops; Boolean/VSM/Python-BM25 use the S-stem indexer.

Table 1: Ranked retrievers on CORD-19 dev, k=10. Boolean AND/OR is unranked; set-level size/recall is in §2.1. Correctness: `tests/test_retrievers.py`.

Run	nDCG@10	MAP@10	MRR	P@10
TF–IDF cosine VSM (`lnc.ltc`, Python)	0.2815	0.0067	0.5178	0.3160
Python BM25, k_1=1.2, b=0.75, δ=0	0.5714	0.0139	0.8393	0.6360
Python BM25⊕VSM fusion (0.85/0.15)	0.5514	0.0129	0.7992	0.6300
C++ BM25, k_1=1.2, b=0.75, δ=0	0.6356	0.0155	0.8875	0.6920
C++ BM25+ (k_1=2.15, b=0.60, δ=0.80), no title	0.6323	0.0151	0.8990	0.6900
+ title β=0.60	0.6815	0.0171	0.9283	0.7380
+ RM3 (α=0.62, μ=9000, 30/40)	0.7458	0.0198	0.9589	0.8260

Table 2: CORD-19, two-process harness. MAP@10 is small because topics have hundreds of relevant documents. Early-phase CORD snapshot from the first leaderboard week (before the current codecs).

Phase	nDCG@10	MAP@10	MRR	Build	Latency	Index
Initial / early CORD run	∼0.735	∼0.019	∼0.93	∼3 s	∼2.9 ms	∼41 MB
This entry (re-measured)	0.7458	0.0198	0.9589	2.20 s	2.28 ms	16.6 MB

Table 3: Same C++ engine on every local collection. **BM25** is textbook Okapi (k_1=1.2, b=0.75, δ=0, no title, no RM3) via `retrieve_bm25_only`. **Ours** is competition `retrieve()` (`adapt_params`). BEIR test splits; Antique test; Vaswani.

Collection	N	avgdl	#q	nDCG@10		MAP@10	P@10	RM3
				BM25	Ours			
Toy	20	7.8	6	0.8793	0.8793	0.8250	0.317	off
Cranfield	1 400	100.8	225	0.2922	0.3308	0.2891	0.271	on
CACM	3 204	33.7	52	0.5020	0.5256	0.2762	0.369	on
Vaswani	11 429	24.8	93	0.4356	0.4571	0.1637	0.381	on
SciFact	5 183	145.5	300	0.6800	0.6832	0.6353	0.090	off
NFCorpus	3 633	156.7	323	0.3240	0.3307	0.1243	0.239	on
FiQA	57 638	71.6	648	0.2476	0.2624	0.1971	0.073	off
SciDocs	25 657	115.2	1 000	0.1568	0.1654	0.0966	0.087	off
ArguAna	8 674	96.5	1 406	0.3690	0.3538	0.2313	0.074	on
Touché-2020	382 545	155.3	49	0.4014	0.4899	0.1543	0.802	off
Quora	522 931	5.8	10 000	0.7858	0.7853	0.7376	0.119	off
Antique	403 666	20.4	200	0.3486	0.4033	0.2279	0.746	off
CORD-19 / TREC-COVID (dev)	171 332	109.7	50	0.6356	0.7458	0.0198	0.826	on

3.1 Parameter search

k_1 and b were re-swept with `retrieve_bm25_only` (RM3 off, no query-length overlay, `title_bonus`=0, δ=0.80) so each curve isolates one knob (Figure 1). On the 0.3-step grid, nDCG rises with k_1 through 2.8 (0.6548); b is flat on $[0.5, 0.8]$ (peak 0.6364 at b=0.70). A dense probe on the same run ($k_1 \in [2.0, 2.4]$, identical b, δ, no title) wobbles by at most 0.006 (e.g. 2.10→0.6337, 2.15→0.6323, 2.18→0.6370): that is 50-topic noise, not a second protocol, so Figure 1 plots only the grid. I treat ±0.01 nDCG@10 as the useful band and do not ship a BM25-only argmax: with RM3 on, k_1=2.4 over-weighted tf on abstracts. Shipped $(k_1, b) = (2.15, 0.60)$ is in the flat b-band and stacks with δ=0.80, β=0.60 (+0.049), then RM3 (+0.064). Moving k_1 from 1.2 to 2.15 *without* title is slightly negative, inside that same band.

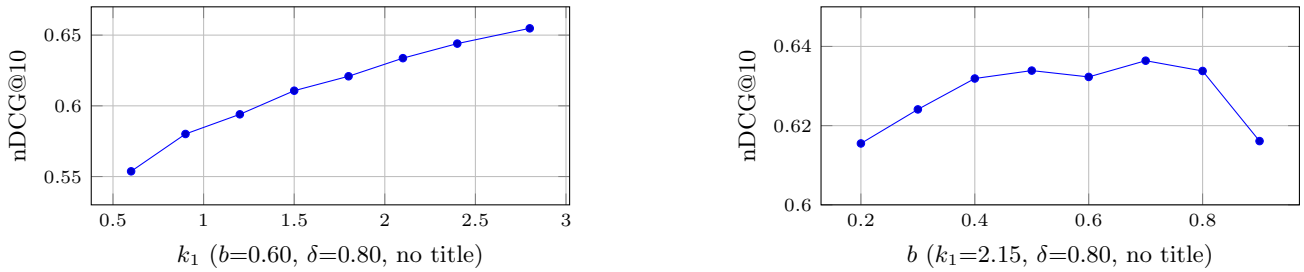


Figure 1: nDCG@10 on the 50 CORD-19 dev topics, BM25+ only, one process. The k_1 panel is the 0.3-step grid only. Off-grid k_1=2.15 is 0.6323, 0.001 below k_1=2.1, inside a ±0.01 50-topic band (dense probe wobble ≤ 0.006). Shipped $(2.15, 0.60)$ is not the BM25-only peak (k_1=2.8).

Overfitting control. The 50-topic (k_1, b) top is flat to within the ±0.01 band above, which is why the shipped pair is a shape function rather than arg max of Figure 1.

4 Error analysis

Five TREC-COVID queries where `retrieve()` is weak (re-measured nDCG@10).

1. **Query 4** (0.066) — *"what causes death from Covid-19?"* Question words are in `INDEX_STOPWORDS`, so the query is mainly {`caus`, `death`, `covid`}. Mortality language is ubiquitous; the mechanism (ARDS) is not in those stems. RM3 expands "death" into more mortality reviews.
2. **Query 12** (0.191) — quarantine best practices. Title-window hits reward early "hospital"/"quarantine" on short overviews. The variance guard only fires on near-constant first-pass scores.
3. **Query 31** (0.266) — coronavirus vs. seasonal flu. Contrastive; bag-of-words cannot require difference. No positions are indexed.

4. **Query 15** (0.366) — live outside the body. Only partly overlaps surface-stability language (query 16, 0.430).
5. **Query 49** (0.481) — long multi-facet NL. Not dump-length, so the overlay leaves RM3 on and the expansion can lock onto one facet.

Query 42 (Vitamin D) is *not* a failure (0.948): `vitamin` is selective after dropping '`d`'. Query 41 ('U.S.') is 0.922.

5 Final competition entry

`retrieve()` is **C++ BM25+ with RM3**, knobs from `adapt_params`(N, avgdl) plus the query-length overlay. It is not the Python fusion (-0.020 nDCG@10 vs. Python BM25). Boolean/VSM remain tested Python components. The bet is shape-based generalisation (Table 3) over a single (k_1, b) fit to 50 topics.

Reproducibility. `README.md`. The extension is compiled in `docker build`, never inside `build_index`. Worker count does not change output bytes; gzip timestamps are zeroed; ties are (score desc, id asc).

AI-use disclosure

LLM assistance (Cursor / Claude) was used for design discussion, code review, and drafting this report. Indexing, codecs, adaptation, and all numbers were implemented and re-measured by me on the course harness.

Code provenance

No search library is used in submitted components. Literature (not a source tree): Okapi BM25 (Robertson & Zaragoza, 2009); BM25+ (Lv & Zhai, 2011); RM3 (Lavrenko & Croft, 2001) with Dirichlet smoothing (Zhai & Lafferty, 2001) — newswire μ=1000, α=0.5, 10+10 started from Anserini and was re-tuned; Simple-9 (Anh & Moffat, 2005); Elias–Fano; front coding and vbyte (Manning et al.). Third-party: `pybind11` and the starter harness. `ir_datasets` is used only to fetch local proxy corpora, not inside `retrieve()`.